

# **OPERATOR'S MANUAL**

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## **CHAPTER FOUR**

### **DAILY OPERATION**



## Introduction

This chapter contains daily turnon, after temporary or long term shutdowns, troubleshooting procedures and an over view of the system triggering / synchronization features.

Refer to Chapter Seven, System Software, for a discussion of the Infinity control software. Refer to Chapter Three, Controls and Indicators, for a discussion of the Infinity optical mounts and their adjustment.




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**Lasing is immediately terminated, and all systems in the laser are sequentially switched off, if an interlock is tripped. To restart the laser close the interlock, clear the Fault Message box, and press the “On” button in the main Infinity control window.**

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**By manually increasing the flash lamp voltage the Infinity system can deliver output energies significantly higher than it is specified to provide. Doing this will, however, permanently damage optical components. Do not run the system at higher energies than what the system is rated for.**

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## Daily Start-Up Procedure

The following procedure should be employed to start the Infinity laser system.

1. Turn the key switch located on the power supply, below the BNC connectors, to **On**.
2. Start the computer and run the Infinity GUI by double clicking on the Infinity icon. After the Infinity library files are bound, the “main” Infinity window should be displayed and the text in the system status panel should read “AC power on”.
3. Select the desired **Repetition rate [Hz]**. This can be done in three different ways.
  - a. Position the mouse cursor over the repetition rate indicator (e.g., downward pointing arrow head in side bar) and depress the left mouse button. This will allow the indicator to be dragged along the slide bar with a resolution of 1 Hz.
  - b. Position the mouse cursor over the side bar to the left or right of the indicator and click the left mouse button. This action will increase or decrease the repetition rate in 10 Hz increments.

- c. Positioning the cursor over the left or right rectangular boxes at the end of the slide bar. Left clicking the mouse at this location will increase or decrease the repetition rate by 0.1 Hz.

Note that as the repetition rate is changed the pulse energy panel is rescaled at 20 and 50 Hz (40-100 systems). Even though it is possible to start the laser at any repetition rate, it is advisable to start at a lower setting in order to minimize thermal stress on components in the experiment (e.g., power meters and optics). The repetition rate can be changed while the laser is running.

4. Select the desired **Pulse energy [mJ]**. Again this can be done in three different ways.
  - a. Position the mouse cursor over the repetition rate indicator (e.g., downward pointing arrow head in side bar) and depress the left mouse button. This will allow the indicator to be dragged along the slide bar with a resolution of 5 mJ/pulse.
  - b. Position the mouse cursor over the side bar to the left or right of the indicator and click the left mouse button. This action will increase or decrease the output power in 25 mJ/pulse increments.
  - c. Positioning the cursor over the left or right rectangular boxes at the end of the slide bar. Left clicking the mouse at this location will increase or decrease the pulse energy by 1 mJ.

The pulse energy can be changed while the laser is running.

5. Press either the **On** or the **Standby** button.

Pressing the **Standby** button will initiate the following sequence of events:

- a. The water pump starts and the water heater is turned on.
- b. The mechanical shutter opens, the master oscillator (DPMO/DPSS) is turned on and the flash lamp is simmered after a 5 second delay.
- c. The "Standby" button is highlighted and Standby is displayed in the status window.

Pressing the **On** button, in addition to the steps listed above, also:

- d. Enables triggering of the flash lamp, and after a ~3 second delay, the Q-switch.

- e. Displays “Laser on” in the status window.

The laser should now be running. If the on-board microprocessor has detected a problem in the system, a new window will appear and an error message will be displayed. Refer to the troubleshooting section of this chapter for more information about error messages.

## **Lamp Voltage Clamp**

As a safety feature you can set the maximum voltage that will be applied to the flash lamp in the main control panel. It is strongly recommended that you set this voltage to only 10 V to 15 V higher than what is needed in order to achieve the desired power. As the lamp ages it can gradually be increased. Be aware that the voltage required to produce 600 mJ/pulse on an old lamp, will when applied to a new lamp, will produce enough energy to destroy optical components.

Under normal operation of the Infinity system the output energy should be set in the pulse energy panel, located in the main control window. The option of manually setting the lamp voltage should only be used for special procedures such as calibration of the internal energy meter.

The LightLoop servo, maintaining long term stability of the laser, is enabled by setting the output energy in the Control window (mJ/pulse). It is disabled by setting the lamp voltage in the Service window rather than the output energy.

## Shut-down Procedure

The lasing action can be terminated by pressing either the **Standby** or the **Off** button in the main user interface.

Pressing the **Standby** button immediately terminates lasing by inhibiting the flash lamp trigger and the Q-switch.

Pressing the **Off** button also switches off the simmer current, the oscillator, the water pump and closes the mechanical shutter. If the computer locks up (i.e., does not respond to mouse or keyboard commands) the laser can be turned off by tripping the external interlock on the power supply. The status of the sub-systems are listed in Table 4-1.

**Table 3-1. System Status in Different Modes**

|                    | <b>OFF<br/>KEY SWITCH ON</b> | <b>STANDBY</b> | <b>ON</b> |
|--------------------|------------------------------|----------------|-----------|
| Lamp               | Not Triggered                | Not Triggered  | Triggered |
| Lamp Simmer        | Not Simmered                 | Simmered       | Simmered  |
| Mechanical Shutter | Closed                       | Open           | Open      |
| Oscillator         | Not Lasing                   | Lasing         | Lasing    |
| Q-Switch           | Not Triggered                | Not Triggered  | Triggered |
| Water Pump&Heater  | Off                          | On             | On        |

## ***Troubleshooting***

This section consists of two parts:

1. A listing of System faults, Table 4-2. System faults are conditions that may exist which make the laser inoperable or cause its performance to be lower than the specifications. How to deal with a specific System fault is described in the respective procedures.
2. A listing of all Fault messages with their accompanying explanations and possible causes and remedies, Table 4-3.

To use the troubleshooting section, find the system fault or fault message that best matches the problem encountered and read the accompanying explanation or remedy to clarify or correct the situation.

For some of the problems listed, the remedy or final step in a procedure may require scheduling a service call. A description of the symptoms and the steps taken attempting to resolve the problem will facilitate the planning of a course of action. The troubleshooting guides may request recording of certain system parameters. Have these at hand when calling Coherent technical support.

***Table 3-2. System Faults***

| <b>SYSTEM FAULTS</b>                       | <b>PROCEDURES</b>     |
|--------------------------------------------|-----------------------|
| System does not turn on                    | Refer to Procedure 1  |
| Low output power                           | Refer to Procedure 2  |
| Unstable laser power                       | Refer to Procedure 3  |
| Depolarized energy high                    | Refer to Procedure 4  |
| Poor spatial mode quality                  | Refer to Procedure 5  |
| Low SHG efficiency                         | Refer to Procedure 6  |
| Low THG efficiency                         | Refer to Procedure 7  |
| Optical breakdown in the SBS liquid        | Refer to Procedure 8  |
| Erroneous IR energy meter reading          | Refer to Procedure 9  |
| Erroneous depolarized energy meter reading | Refer to Procedure 10 |

**Table 3-3. Fault Messages**

| MESSAGE                                                   | EXPLANATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\pm 12$ VA, $\pm 12$ VB, +2 VB, +4 VB, +5 V out of range | The listed supply voltage has fallen outside its normal operating range. This message can sometimes appear as transients disturb the system. If the system otherwise behaves normal clear the message and continue to operate it. If the message reappears check that the line voltages on all three phases are within their specifications. If they are, call Coherent's technical support.                                                                                                                                                                                                                                                                                     |
| AC line voltage low                                       | The voltage on at least one of the incoming lines is too low to ensure the proper and safe operation of the system. Check all incoming phases against neutral.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Boost board fan not turning                               | The fan cooling the booster boards does not work. If you continue to run the system, the booster boards will overheat and eventually fail catastrophically. If this message appears call Coherent's technical support.                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Cap bank not charged                                      | At the end of the charge cycle, the voltage across the capacitor bank does not equal the value set by the computer. If this message appears call Coherent's technical support.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Contactor fan not turning                                 | The fan cooling the two boards supplying low voltages ( $\pm 12$ VA, $\pm 12$ VB, +2 VB, +4 VB, +5 V) to the system electronics does not work. If you continue to run the system the boards will overheat and eventually fail catastrophically. If this message appears call Coherent's technical support.                                                                                                                                                                                                                                                                                                                                                                       |
| Control system failure                                    | The system status as read by the computer is not the same as that set by the computer. Consequently the computer may not have full control over the system. When this fault occurs, the CPU attempts to shut down the entire system. However, owing to the nature of this fault there is no guarantee that this will be successful. If this message appears turn off the system if it is still running. Disconnect any test equipment that may be connected to it. Remove possible sources of EMI from the lasers vicinity. Disconnect all BNC cables from it. Make sure that the PC computer is working. If this message continues to appear call Coherent's technical support. |
| Depolarization too high                                   | The depolarized energy returning from the amplifier is too high. Since this energy goes back into the DPSS laser it may jeopardize the safe and optimum performance of the master oscillator. See Procedure 4 for the depolarized energy minimization procedure.                                                                                                                                                                                                                                                                                                                                                                                                                 |
| DPSS electronic fault                                     | <p>The diode pumped master oscillator (DPSS) is stabilized by several servo loops. If one or more of these fail to keep the system parameters within their operating ranges, this message is displayed.</p> <p>Possible causes are overheating and electromagnetic interference.</p> <p>There are no serviceable parts inside the DPSS housing. If this message appears repeatedly call Coherent's technical support.</p>                                                                                                                                                                                                                                                        |



| MESSAGE                        | EXPLANATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EPROM hardware fault           | The EEPROM in the power supply does not respond to commands from the PC computer. If this message appears call Coherent's technical support.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| External (CDRH) interlock open | The Infinity system is equipped with an external interlock circuit that can be used to monitor the status of external devices. It is located at the rear of the power supply. If this circuit is open the system will not start. The system can be restarted by clicking on ON in the user interface as soon as the interlock circuit has been short circuited. See Chapter 5 for a description of how to install an external safety switch.                                                                                                                                                                                             |
| FlashTrack at limit            | The FlashTrack servo system has, at least temporarily, been unable to perform its task. Verify that the cable is connected to the photo diode housing.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| High water temperature         | <p>The temperature of the internal cooling water exceeds 45°C. All subsystems in the laser are switched off.</p> <p>Make sure that the water pump is working and that the incoming external cooling water is colder than 30°C. Check that the incoming water can flow freely through the hoses and the umbilical.</p> <p>If the problem persists, remove the water side cover on the power supply. Turn the regulating valve, located between the heat exchanger and the deionization filter, see Chapter 3, half a turn clockwise.</p> <p>Restart the water pump (from the service window) when the internal water has cooled down.</p> |
| Interlock open                 | <p>There are five interlocks in the system. They are: 1. Laser head cover interlock, 2. Flash lamp terminal box cover, 3. External interlock and 4&amp;5, the two interlock dowel pins next to the latches holding the POT. If any of these interlocks are tripped all systems in the laser are switched off.</p> <p>In order to restart the system close the interlock circuit and click on the ON button in the main control window.</p>                                                                                                                                                                                               |
| Lamp not simmering             | The system has failed in igniting the lamp. When the lamp is simmered a steady whitish glow can be seen from the rods. Check that the lamp leads are connected and that the lamp itself is undamaged. Try to simmer the lamp again by first turning the laser off and then on again. If the lamp does not simmer after a couple of attempts, call Coherent's technical support.                                                                                                                                                                                                                                                          |
| Low capacitor bank voltage     | If the voltage across the capacitor bank falls below the line voltage this error message is displayed and the system is turned off. The voltage regulation circuits can only function above the line voltage. If the voltage across the capacitors is lower at the beginning of the charge cycle it will behave as a short circuit until the voltage is brought up to the line voltage. In order to avoid the sudden rush of current into the system that would occur, the capacitor bank voltage is monitored. If this message cannot be defeated by turning the laser off and restarting it, call Coherent's technical support.        |

| MESSAGE                                                                                                                                                                                          | EXPLANATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Low water flow                                                                                                                                                                                   | The flow rate in the internal cooling loop is too low to ensure safe operation of the system. Remove the water side cover on the power supply. Check that there are no leaks and that the particle filter is in good shape. Remove the cap on the end of the pump and by switching the pump on and off establish that it is spinning clockwise.                                                                                                                                                             |
| Power module A over current<br>Power module A over temperature<br>Power module A voltage fault<br>Power module B over current<br>Power module B over temperature<br>Power module B voltage fault | These messages indicate that one or both of the booster boards have failed. If these messages cannot be defeated by turning the laser off and restarting it, call Coherent's technical support.                                                                                                                                                                                                                                                                                                             |
| Remote buffer overflow                                                                                                                                                                           | The PC sent a message to the power supply before the previous message was acknowledged by the power supply. Since the message exchange is under software control, this error would normally indicate a software defect, although it could possibly be caused by a noisy serial line or defective serial port. Check that the serial port and the cable between the computer and power supply is OK. Eliminate any possible sources of EMI. If possible try another computer.                                |
| Remote framing error                                                                                                                                                                             | The UART on the CPU reported a framing error, which typically results from a baud rate mismatch between the serial sender and receiver. Since the baud rate settings are under software control, this error would normally indicate a software defect, although it could possibly be caused by a noisy serial line or defective serial port. Check that the serial port and the cable between the computer and power supply is OK. Eliminate any possible sources of EMI. If possible try another computer. |
| Remote parity error                                                                                                                                                                              | The UART on the CPU board reported a parity error indicating noise on the serial line. Check that the serial port and the cable between the computer and power supply is OK. Eliminate any possible sources of EMI. If possible try another computer.                                                                                                                                                                                                                                                       |
| Remote UART overrun                                                                                                                                                                              | The UART on the CPU reported a character overrun, a condition in which a character was received while the previous character was still read by the CPU. This indicates a software defect. Reinstall the software.                                                                                                                                                                                                                                                                                           |
| SCR failure                                                                                                                                                                                      | The Infinity system has two solid state rectifiers. If one or both of them fails, this message is displayed. Call Coherent's technical support.                                                                                                                                                                                                                                                                                                                                                             |
| Servo error: no lasing                                                                                                                                                                           | If the internal power meter reads less than 10 mJ/pulse and the lamp voltage is above 460 V this message is displayed. Check that there are no obstructions in the beam path and that the cable to the IR photodiode housing is connected, see Chapter Three.                                                                                                                                                                                                                                               |

| MESSAGE                     | EXPLANATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Transformer fan not turning | The fan cooling the transformer does not work. If the system does not have a built in transformer the system can be safely operated without the fan working (only 380 V systems have this transformer). Call Coherent's technical support for further assistance.                                                                                                                                                                                                                                                                                                                                               |
| Water flow low              | See low water flow                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Water level low             | The water level in the reservoir, located in the power supply, is to low.<br>Refill with deionized water and check for leaks.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Water resistivity low       | The resistivity of the internal cooling water is to low.<br>Remove the cover on the water side on the power supply. Turn the bypass valve, see Chapter 3 for location, half a turn counter clockwise. Wait until the resistivity, value available in the Monitor window, stabilizes. Repeat if necessary. The recommended value is 3 M $\Omega$ /cm and the recommended range 2 – 6 M $\Omega$ /cm. The laser can not be operated when the resistivity is lower than 1.5 M $\Omega$ /cm. Replace the deionization cartridge when the resistivity, even with the valve fully open, stays below 2 M $\Omega$ /cm. |
| Water resistivity high      | The resistivity of the internal cooling water is to high.<br>Remove the cover on the water side on the power supply. Turn the bypass valve, see Chapter 3 for location, half a turn clockwise. Wait until the resistivity, value available in the Monitor window, stabilizes. Repeat if necessary. The recommended value is 3 M $\Omega$ /cm and the recommended range 2 – 6 M $\Omega$ /cm.                                                                                                                                                                                                                    |
| Wrong AC phase rotation     | This means that the relative order of the three electrical phases are wrong.<br>Interchange any two of the incoming AC phases and try again.                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

## **Procedure 1. System fault: Laser does not start**

When the power supply is turned on, a 0 should be displayed on the readout above the BNC connectors and the fans inside the power supply and laser head should come on. If these signs are absent check the following:

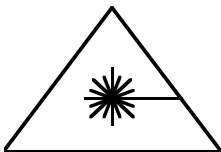
1. That the facility main power circuit breaker is turned on.
2. That the line voltage is correct and that all three phases are active.
3. That none of the three line fuses, located at the rear of the power supply, are blown.
4. That the key switch on the power supply is turned all the way to the ON position.

If the power supply does not start, schedule a service call.

## Procedure 2. System fault: Low output power

Note that the internal power meter only monitors the IR energy. If the energy harmonic energy is low see Procedures 6 through 8. The procedure below assumes that the energy at 1064 nm is low. Low output power can be due to a number of factors. Check the following:

1. Confirm that the laser power is low using an external calibrated power meter. If necessary recalibrate the built-in energy meter, see Procedure 9.
2. That there are no obstructions in the beam path and that the entire beam goes into the power meter. Verify that the beam is not being clipped anywhere by taking a burn pattern of the mode just after the last folding mirror. It should appear round and symmetrical. If the burn mark exhibits any irregularities or asymmetries, the laser might be misaligned or one or more of its optical components damaged. See Chapter Six, Maintenance, for optics inspection procedures.
3. That the maximum lamp voltage in the Infinity Control window is at least 10 V higher than what is needed.
4. The number of shots the flash lamp has fired. A shot counter is available in the Infinity Control panel. If this number exceeds 75 million shots, change the flash lamp. The flash lamp replacement procedure is described in Chapter Six, Maintenance.
5. That the flash lamp and Q-switch timing sequence is optimized as well as the emission time of the flash lamp. In the Infinity Control panel click on the Service button and then chose first Lamp from the menu bar and then Timing from the menu. The Lamp pulse width should be close to 220  $\mu$ s and the QS trigger position about 250  $\mu$ s. These delays are factory optimized and there is, under normal operating conditions, no reason for the user to change them.



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**The procedure described below requires the laser to be operated with the laser head cover open and the interlock defeated. It is strongly recommended that everyone present in the room wears appropriate safety glasses and that access to the room is restricted.**

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**Even though positioning the Q-switch trigger position later with respect to the lamp pulse than indicated above, may yield higher pulse energies it will also lead to instabilities in the laser mode and intensity. It is imperative that the Q-switch trigger is positioned before the turn off time of the flash lamp.**

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6. That the output power from the oscillator is at least 20  $\mu\text{J}$ /pulse. While performing this measurement be aware of the cw prelasing radiation emitted from the oscillator. This radiation amounts to  $\sim 15$  mW of power while the integrated Q-switched output at 100 Hz repetition rate equals 3 mW (100 Hz\*30 $\mu\text{J}$ /pulse). It is therefore necessary to measure the pulsed output with a calibrated photo-diode, a Gentec energy meter or similar pulse sensitive meter. If the output energy is below 20  $\mu\text{J}$ /pulse schedule a service call.

### Procedure 3. System fault: Unstable laser power

The shot-to-shot intensity fluctuations are specified to have a standard deviation of less than 1.0 % for repetition rates up to 50 Hz at maximum system pulse energy (600 mJ/pulse up to 20 Hz, and 500 mJ/pulse from 20.1 up to 50 Hz repetition rate). At repetition rates above 50 Hz the specified standard deviation is 1.7 % for an output of 400 mJ. If the output at 1064 nm exhibits larger intensity variations, check the following

1. Verify that the laser is being operated at full power. Running the laser at lower pulse energies will inevitably lead to increased intensity fluctuations. This is especially true at energies close to the SBS threshold energy.



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**Attenuating the master oscillator output or moving the Q-switch trigger with respect to the flash lamp emission will reduce the output power, but it will also increase the standard deviation of the shot to shot noise!**

**To achieve a combination of low noise and low power the beam should be attenuated outside the laser using neutral density filters or glass wedges.**

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2. Verify that FlashTrack is active by opening the “Servo” window. FlashTrack is an electronic control servo that, based on the integrated lamp emission during the first half of the pulse, makes a small correction to the current going into it during the second half to keep the total amplification constant.
3. The number of shots the flash lamp has fired. A shot counter is available in the Infinity Control panel. If this number exceeds 75 million shots, change the flash lamp. The flash lamp replacement procedure is described in Chapter Six, Maintenance.
4. That the flash lamp and Q-switch timing sequence is optimized as well as the emission time of the flash lamp. These parameters are set in the Infinity lamp timing window that is accessed from the Lamp menu in the Service window. Verify that the parameter values, Lamp pulse width and Q-switch delay, match those on the performance sheet delivered with the system. These delays are factory optimized and there are, under normal operating conditions, no reasons for the user to change them.



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**Even though positioning the Q-switch trigger position later with respect to the lamp pulse than indicated above, may yield higher pulse energies it will also lead to instabilities in the laser mode and intensity. It is imperative that the Q-switch trigger is positioned before the turn off time of the flash lamp.**

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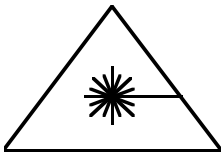
5. The DPSS temperature. If the oscillator runs in a cavity mode that is close to an adjacent mode, occasional mode-jumps might occur. Multi-mode output will lead to breakdown in the SBS cell, which can be heard as a distinct clicking sound. If this is present a small change in the oscillator temperature set-point is usually enough to remedy the problem. To do this, first in the Infinity Control panel click on the “Service” button, secondly in the Infinity Diagnostic panel move the DPSS heater slide bar, located in the lower left corner, 5 counts in either direction. Allow a few minutes for the oscillator to reach its new equilibrium temperature. If necessary repeat a couple of times until the sound disappears.
6. Verify that there are no intense emitters of electromagnetic interference close to the Infinity system. Examples of such emitters are Excimer lasers and radio frequency devices.



## Procedure 4. System fault: Depolarized Energy High

Depolarized energy is defined as all radiation which is not vertically polarized after transversing the Infinity amplifier stage two times. Since this radiation is not vertically polarized, it is not coupled out of the amplifier by the thin film polarizer. It is instead transmitted through the thin film polarizer and directed back into the Faraday isolator.

It is desirable to keep this energy component to no more than 1.5% of the output energy, when operating the laser at its maximum energy and repetition rate. If the energy going through the Faraday isolator, as measured by the photo diode attached to it, exceeds 2 % of the output energy, the on-board microprocessor will shut off the system. This is done in order to protect the master oscillator from being damaged. A bar graph display showing the depolarized energy is available in the Infinity software Monitor window which can be accessed by clicking on the “Monitor...” button in the main Infinity control window.



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**The procedure described below requires the laser to be operated with the laser head cover open and the interlock defeated. It is strongly recommended that everyone present in the room wears appropriate safety glasses and that access to the room is restricted.**

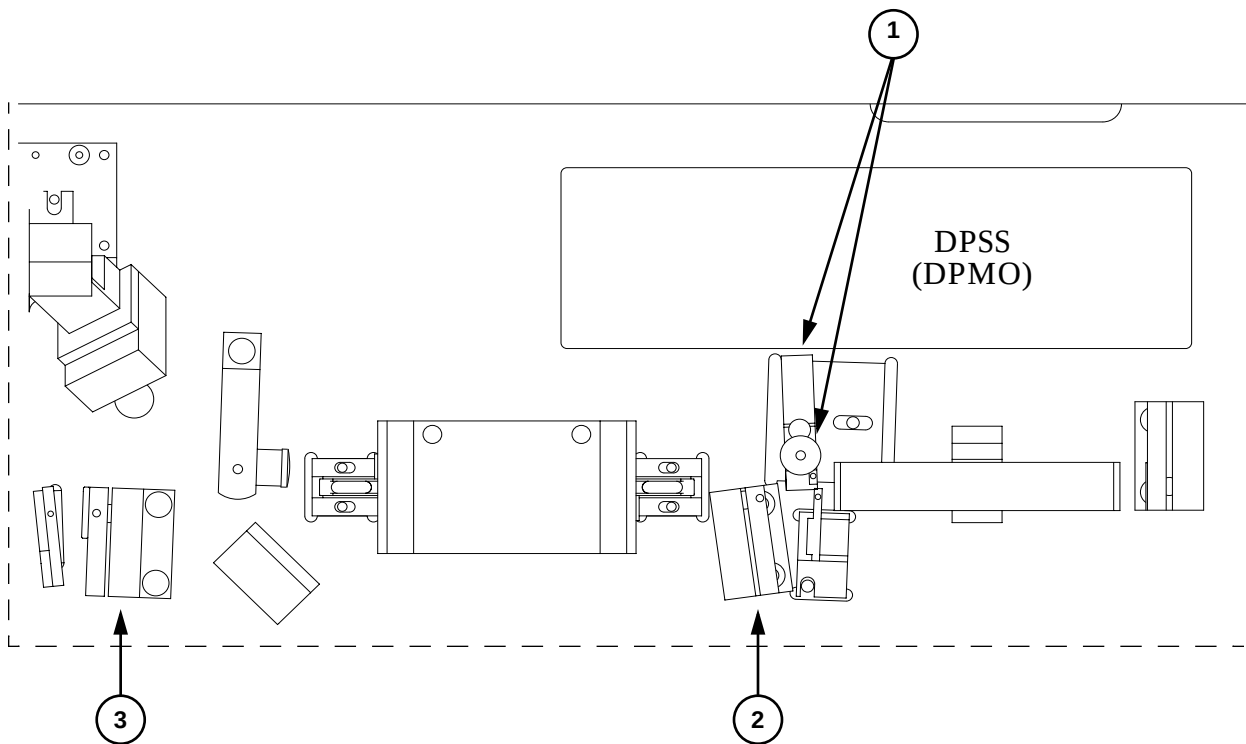
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Follow the procedure described below to minimize the amount of depolarized energy.

1. Increase the repetition rate to the maximum value for the system, and thereafter the IR energy until the depolarized energy is 1% of the maximum output power. If the depolarized energy falls below 1%, as a result of the adjustments described below, increase the IR pulse energy until 1% is again reached. Continue to increase the IR energy until the maximum pulse energy of the system is reached.
2. Slowly adjust the horizontal and vertical translation controls of the first relay lens, mount 1 in Figure 4-1, while monitoring the amount of depolarized light. Iterate between the two controls until the minimum is reached.
3. Adjust the vertical and horizontal tilt angle of the 90° polarization rotator, mount 2 in Figure 4-1. Use the long 5/32” ball driver supplied with the laser. Slowly tilt

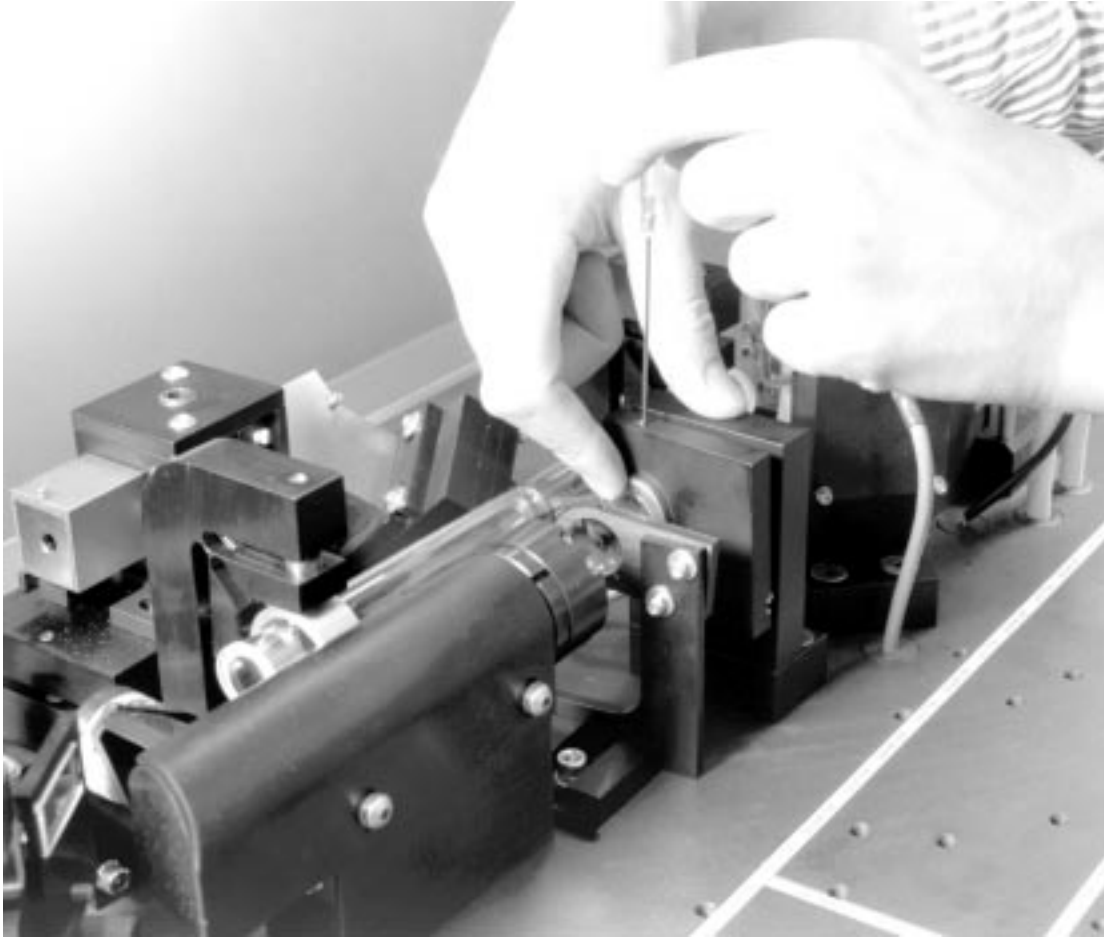
the holder in the vertical and horizontal direction back and forth until a minimum in the depolarized energy is found. Iterate between the two directions.

4. Make small vertical and horizontal tilt adjustments of the quarter wave plate holder, mount 3 in Figure 4-1, until the amount of depolarized energy has been minimized.



**Figure 3-1. Order of Adjustments for Minimization of Depolarized Energy**

5. Reduce the power to its minimum value and then reduce the repetition rate to 10 Hz. Slowly increase the power until the depolarized energy reaches 5 mJ or until the maximum pulse energy of the system is reached, 600 mJ/pulse.
6. While holding the quarter wave plate holder in place with a finger, loosen the setscrew holding the quarter wave plate as shown in Figure 4-2. Carefully, while monitoring the intensity of the depolarized light, make small rotational adjustments until the amount of depolarized energy has been minimized.
7. Repeat steps 5 through 6 until no further decrease in the depolarized energy can be obtained or the maximum pulse energy of the system is reached.



**Figure 3-2. Alignment of the Quarter Wave Plate**

8. Repeat steps 1-7 until it is possible to run the system at its maximum average output power (i.e., maximum energy and pulse rate) with less than 1.5% depolarized energy.

If the depolarized energy can't be brought below 1.5% call Coherent's technical support group.

## Procedure 5. System fault: Poor Mode and Beam Quality

Poor mode quality means that the beam exhibits an asymmetrical intensity distribution, for example ellipticity or hot spots, while poor beam quality means that the beam is more divergent than what is specified for the system. They are related in so far that a poor mode usually implies poor beam quality. The reverse is, however, not necessarily true. A beams intensity distribution can appear symmetrical and it can still be highly diverging. In order to achieve the very high beam quality that the Infinity system is capable of delivering it is important that the beam is round and symmetrical.

The mode quality can be checked using burn paper or, preferably, a CCD camera based beam analyzing system, such as the Coherent BeamView Analyzer.

The Infinity system has the following two specifications for its beam quality. In the near field 95% of the energy can be transmitted through an aperture with a diameter of 5.5 mm. In the far field the beam will have a divergence no greater than 0.7 mrad full angle, measured between points defining an area containing 84% of the total energy.

If the IR beam appears asymmetrical check the following:

1. That the laser is operated at full power
2. That the mode is diagnosed in the near field. The near field is located about 5 cm after the thin film polarizer.
3. That there are no obstructions in the beam path and that the entire beam strikes the burn paper or the CCD camera. In particular make sure that the white teflon shields protecting the flash lamp leads are pushed all the way down into their mounts.
4. All optical components including the rod surfaces (see Chapter Six for optics inspection procedures and cleaning). If any of the optics are found to be damaged schedule a service call.



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**Do not try to improve the mode quality by modifying the alignment. Doing this may damage the optics in the laser.**

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If the IR beam exhibits a larger diameter in the near field or greater divergence in the far field than specified, verify the following:

1. That the collimating lens, positioned after the Faraday isolator, has not been moved.
2. That the beam is not striking the pinhole in the vacuum spatial filter. This can be checked with an IR viewer. A consequence of the amplified return beam striking the pinhole is the development of a dark discoloration on the inside of the glass envelope. In extreme cases it might be possible to see a bluish plasma on the pinhole on the POT side. If this is the case, turn off the system and call Coherent's technical support group.



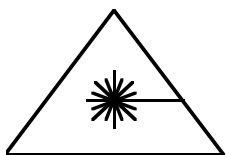
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**It should be noted that a smoky discoloration of the spatial filter vacuum cell will occur during normal operation. An indication that the filter is misaligned would be a deep blackening of the glass vacuum cell.**

---

## Procedure 6. System fault: Low Second Harmonic Generation Efficiency

Second Harmonic Generation, or SHG for short, is an optical non-linear process in which two incoming photons interact with a medium to form a third new photon. Since energy is conserved, the new photon has an energy equal to the two incoming ones. This means that the efficiency of the process depends mainly on the intensity of the incoming radiation field. However, there are two other important factors which must be considered when determining the efficiency of the SHG process. These are beam quality (its brightness) and the quality of the crystal used. It is therefore only meaningful to talk about efficiency for a given beam / crystal configuration.



---

**Some of the steps in the procedure below require the head cover to be opened and the head interlock defeated. Use appropriate safety glasses and restrict access to the room in which the laser is operated to qualified personnel.**

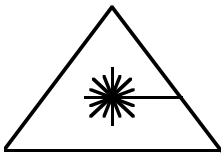
---

When the Infinity system is operated at 600 mJ / pulse the SHG efficiency should be higher than 50%. If this is not the case check the following:

1. That the crystal is optimally phase-matched. While monitoring the laser power at 532 nm slowly turn, using a 9/64" allen wrench, the horizontal adjustment screw on the SHG mount until maximum power is achieved.
2. That the IR energy is 500 mJ/pulse using an external power meter.
3. That the mode appears round and symmetrical. Use the burn paper supplied with the laser.
4. That the BBO crystal and the cell windows are undamaged. Remove the cell by first unscrewing the two lock nuts, then remove the plate and finally pull the crystal housing out of the mount. Inspect the cell for damages, burn marks, discolorations etc. If any damage is found on the windows or the crystal call your Coherent representative.
5. That the harmonic mount temperature remains stabilized (i.e., remains in the hold configuration) during operation. See Chapter Seven, System Software, for more information on this window.

## Procedure 7. System fault: Low Third Harmonic Generation Efficiency

Third Harmonic Generation, or THG for short, in a Nd:YAG laser, is an optical non-linear process in which the radiation fields at 1064 nm and 532 nm interact in a non-linear medium to form a third field radiating at 355 nm. The efficiency of the process depends on the intensity of the beams, their spatial and temporal mode and, the quality of the crystal.



---

**Some of the steps in the procedure below require the head cover to be opened and the head interlock defeated. Use appropriate safety glasses and restrict access to the room in which the laser is operated to qualified personnel.**

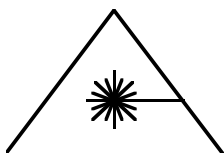
---

If the IR energy is 600 mJ/pulse and the SHG energy is greater than 250 mJ/pulse, the energy at 355 nm should be higher than 200 mJ/pulse. If this is not the case check the following:

1. That both crystals are optimally phase-matched. While monitoring the laser power at 355 nm slowly turn, using a 9/64" allen wrench, the vertical adjustment screw on first the THG mount until maximum power is achieved and then the horizontal adjustment screw on the SHG mount. Note, the optimum phase-matching angle for generation of 532 nm is different when generating only 532 nm and when generating it as an intermediate step for 355 nm generation.
2. That the IR energy is 600 mJ/pulse and that the energy at 532 nm is above 250 mJ/pulse using an external calibrated power meter.
3. That the waveplate between the SHG and THG crystals is rotated to its optimum position.
4. That the mode appears round and symmetrical. Use the burn paper supplied with the laser.
5. That the BBO crystals and the cell windows are undamaged. If any damage is found on the windows or the crystals call Coherent.
6. That the harmonic mount temperature remains stabilized (i.e., remains in the hold configuration) during operation. See Chapter Seven, System Software, for more information on this window.

## Procedure 8. System fault: Low Fourth Harmonic Generation Efficiency

Fourth Harmonic Generation, or FHG for short, is an optical non-linear process in which two incoming photons interact with a medium to form a third new photon. Since energy is conserved, the new photon has an energy equal to the two incoming ones. This means that the efficiency of the process depends mainly on the intensity of the incoming radiation field. However, there are two other important factors which must be considered when determining the efficiency of the SHG process. These are beam quality (its brightness) and the quality of the crystal used. It is therefore only meaningful to talk about efficiency for a given beam / crystal configuration.



---

**Some of the steps in the procedure below require the head cover to be opened and the head interlock defeated. Use appropriate safety glasses and restrict access to the room in which the laser is operated to qualified personnel.**

---

With the repetition rate set to 10 Hz and a SHG energy equal to or greater than 250 mJ/pulse, the energy at 266 nm should be greater than or equal to 50 mJ/pulse. If this is not the case check the following:

1. That both crystals are optimally phase-matched. While monitoring the laser power at 266 nm slowly turn, using a 9/64" allen wrench, the vertical adjustment screw on FHG mount until maximum power is achieved and then the horizontal adjustment screw on the SHG mount.
2. That the IR energy is 600 mJ/pulse and that the energy at 532 nm is above 250 mJ/pulse using an external calibrated power meter.
3. That the SHG beam is cleanly passing through the strip out dichroic and that the back reflection from the FHG cell is also cleanly passing through the strip out dichroic and striking the beam dump (white ceramic diffuser) mounted on the SHG stage.
4. That the mode appears round and symmetrical (A white business card works well for this).
5. That the BBO crystals and the cell windows are undamaged. If any damage is found on the windows or the crystals call Coherent.



6. That the harmonic mount temperature remains stabilized (i.e., remains in the hold configuration) during operation. See Chapter Seven, System Software, for more information on this window.



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**Due to the large nonlinear coefficient of BBO at 266 nm, 133 nm radiation can also be generated by the same crystal employed to generate the 266 nm radiation. BBO is not transparent to 133 nm radiation resulting in a significant temperature gradient as the pump energy and repetition rate of the laser system are changed. It is therefore necessary, and normal, to have to adjust the phase matching angle of the FHG crystal if the laser system operational parameters are changed by more than a few percent.**

---

## **Procedure 9. System fault: Breakdown in SBS Cell**

If the oscillator runs in a cavity mode that is close to an adjacent mode, occasional mode-jumps may occur. Multi-mode output will lead to laser induced breakdown in the SBS cell, which can be heard as a distinct clicking sound. If this is present, a small change in the oscillator temperature is usually enough to remedy the problem. To do this, first in the Infinity Control panel click on the “Service” button, secondly in the Infinity Diagnostic panel move the DPSS heater slide bar, located in the lower left corner, 5 counts in either direction. Allow a few minutes for the oscillator to reach its new equilibrium temperature. If necessary repeat a couple of times until the sound disappears.

If no improvement can be achieved call Coherent Technical Support.

## **Procedure 10. System fault: Erroneous IR Energy Meter Reading**

It is possible to recalibrate the built in energy meter through the system software. Since some parts of the energy meter system are temperature sensitive allow the laser to reach thermal equilibrium before evaluating and calibrating it. Run the laser at its maximum repetition rate and pulse energy for one hour with the cover on (the average power should be measured on an external power meter!). If the energy meter still reads incorrectly, follow the procedure below. Numbers in the text are given with reference to Figure 4-3.

1. In the main Infinity control window click on “Service...” and then chose “IR...” from the Service window Calibration pull down menu.
2. The window shown below, Figure 4-3, should now be displayed on the screen. Move the Calibration and the Service windows on the screen so they do not totally obscure each other.
3. Optimize the signal on the detector. Note that the “Lamp volts” slide bar in the Service window should be used to set the IR pulse energy of the laser system during this procedure.

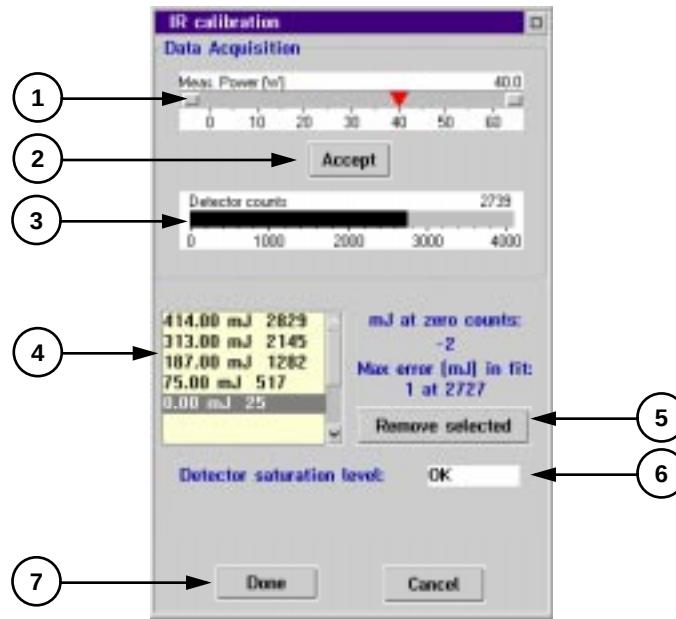
For 40-100 systems, set the repetition rate to 100 Hz and the output energy to 200 mJ using an external power meter.

For 25-50 systems, set the repetition rate to 50 Hz and the output energy to 250 mJ using an external power meter.

For 15-30 systems, set the repetition rate to 30 Hz and the output energy to 250 mJ using an external power meter.

Loosen the screws which hold the photodiode housing to the fold optic assembly. While watching the “Detector counts” bar graph, move the photodiode housing until a maximum is found, see Figure 4-3.

4. Set the lamp voltage in the service window so that the external energy meter reads the maximum specified average power of the system.
5. Set the power measured on the energy meter on the slide bar in the calibration window, item 1.
6. When the detector counts value, item 3, has stabilized, click on Accept, item 2. The entered value will appear in the window, item 4.
7. In the Service window decrease the lamp voltage by approximately 15 volts.



**Figure 3-3. IR Energy Calibration Window**

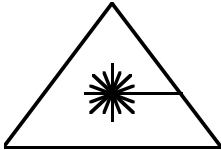
8. Repeat steps 5 to 7 at least 3 times. The exact value of the entered powers does not matter but it is important that 0 mJ be the last data point entered. Record this data point with the lamp voltage set lower than the lasing threshold. (i.e., 360 V).
9. Compute a new set of interpolation constants by clicking on "Done", item 7.
10. The IR energy meter is now calibrated. Go back to the main control window.

If an incorrect value is entered you can removed it by first highlighting it and then click on "Remove selected", item 5.

If there is not enough, or too much light, on the photodetector a message will be displayed stating in which direction the light level needs to be adjusted, item 6. The detector adjustment should be made at the current pulse energy (i.e., lamp voltage). For example; if after entering the second data point the system indicates that the detector counts need to be increase by 10%, and the current number of counts is 2000, then the detector counts should be increased to 2200 with the system operating at the lamp voltage setting of data point number two. After adjusting the detector counts, all data points should be removed and the calibration procedure restarted from step 4.

## Procedure 11. System fault: Erroneous Depolarized Energy Meter Reading

The photodiode monitoring the depolarized energy can be recalibrated through the system software. Follow the procedure below. It is assumed that the internal IR energy meter is calibrated. If not, the IR detector should be calibrated first, see Procedure 10.



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**The procedure described below requires the laser to be operated with the laser head cover open and the interlock defeated. It is strongly recommended that everyone present in the room wears appropriate safety glasses and that access to the room is restricted.**

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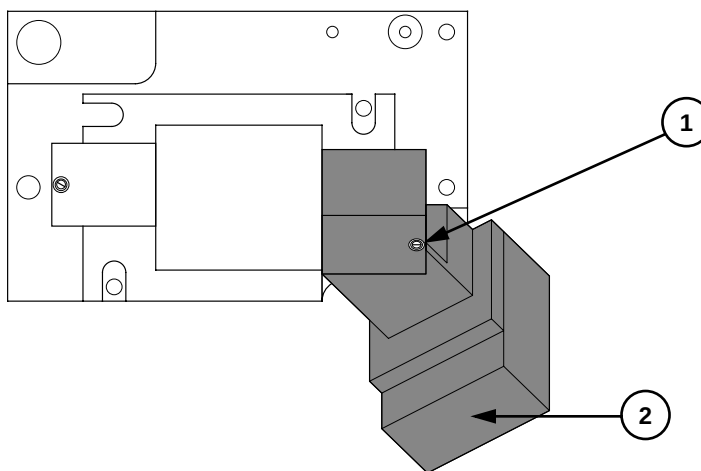
In order to access the depolarized beam, the photodiode attached to the Faraday isolator will need to be removed. Numbers in the text are given with reference to Figure 4-4.

1. Open the head cover and install the head interlock defeat. Disconnect the electrical plug on the back of the photodiode assembly, item 2. Loosen the retaining screw, item 1, and pull the diode assembly (grey in figure) up and out of the laser. Be careful not to touch the spatial filter.
2. Position the Infinity beam dump, or an external power meter, to intercept the main IR beam.
3. In the main Infinity control window set the output energy to 0 mJ and the repetition rate to 10 Hz. Start the laser by clicking on “On”.
4. In the main Infinity control window click on “Service...” and then chose “Depolarization...” from the Service window Calibration pull down menu.
5. The window shown below, Figure 4-5, should now be displayed on the screen. Move the Calibration and the Service windows on the screen so they do not totally obscure each other.
6. Gradually increase the lamp voltage until depolarized light can be detected using an IR fluorescent card. This light will be directed towards the Infinity baseplate in the direction of the Spatial Filter.

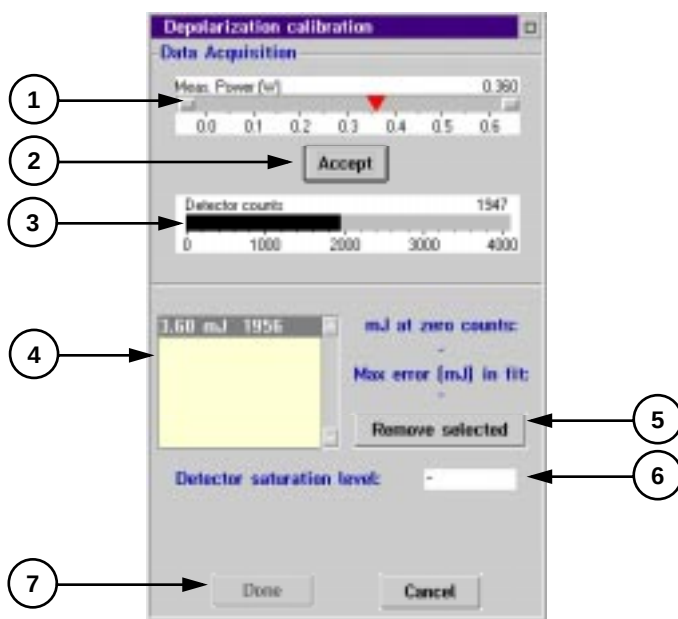
7. Position a IR folding mirror so that the beam reflected off the exposed polarizer is directed into an external power meter. Make sure, using the IR viewing card, that the beam goes into the power meter unobstructed.



**It is imperative that a reliable and calibrated power meter is used for this procedure and that it is carried out in a meticulous way. Erroneous calibration of the depolarized energy monitor may lead to irreparable damage.**



**Figure 3-4. Faraday Isolator and Photo-Diode Assembly Measuring Depolarized Energy**



**Figure 3-5. Depolarized Energy Calibration Window**

8. Set the lamp voltage, in the Service window, to 360 V and then close the Service Window. Increase the repetition rate to system maximum. Let the system warm up by running it for >15 minutes.
9. Collect data points, depolarized energy vs. output energy, according to the following procedure. Note that the software will want the depolarized energy readings to be in “watts”.
  - a. Record the internal energy meter value, available in the main Infinity control window, and the corresponding depolarized energy on the external power meter. Take the first reading at 360 V (i.e., 0 mJ output).
  - b. Set the IR energy to 100 mJ.
  - c. Take a reading of the depolarized energy.
  - d. Increase the IR energy by 50 mJ.
  - e. Repeat steps “c” and “d” until the maximum output energy of the system is reached.
10. Set the IR energy to 1 mJ, and place the laser system in “Standby”. Reinstall the photodiode assembly.
11. Turn the laser system back on and set the IR energy to the system maximum. Loosen the screws which hold the photodiode housing to the fold optic assembly. While watching the “Detector counts” bar graph, move the photodiode housing until a maximum is found, see Figure 4-5.
12. Set the “Meas. Power” slide bar, item 1, to the measured depolarized energy value.
13. When the detector counts value, item 3, has stabilized, click on Accept, item 2. The entered value will appear in the window, item 4.
14. Set the IR energy to the value used for the next depolarized energy data point.
15. Repeat steps 12 through 14 for each of the depolarized energy values recorded. Note that Service window “Lamp volts” slide bar will need to be used for the last data point (i.e., the 0 mJ data point at a lamp voltage of 360 V).
16. Compute a new set of interpolation constants by clicking on “Done”, item 7.

17. The depolarized energy meter is now calibrated. Go back to the main control window.

If an incorrect value is entered you can remove it by first highlighting it and then click on “Remove selected”, item 5.

If there is not enough, or too much light, on the photodetector a message will be displayed stating in which direction the light level needs to be adjusted, item 6. The detector adjustment should be made at the current IR pulse energy. For example; if after entering the second data point the system indicates that the detector counts need to be increased by 10%, and the current number of counts is 200, then the detector counts should be increased to 220 with the system operating at the IR pulse energy of data point number two. After adjusting the detector counts, all data points should be removed and the calibration procedure restarted from step 11.



## **Timing and Synchronization**

Infinity's on-board, digital timing and triggering system provides a powerful set of tools for synchronizing Infinity with external laboratory equipment. Several input triggers and output sync signals are available, allowing Infinity to operate as either a master or a slave. All interface signals use TTL logic, the most common and convenient standard for laboratory instruments. All timing parameters are under control of the Infinity CPU and are therefore accessible through any of Infinity's user interfaces. The system includes advanced features designed to facilitate laboratory automation, such as programmable delay generators and sync signals. In some applications, Infinity's built-in capabilities will eliminate the need for external delay generators.

The material in this section can be divided into two topics: triggering Infinity from external instrumentation, and synchronizing external instrumentation to Infinity. These two topics are concerned with Infinity's inputs and outputs, respectively.

*Inputs* - Infinity can respond to two sets of input signals: (1) external triggers that allow direct control of repetition rate and pulse timing; and (2) blanking inputs that allow laser emission to be inhibited by an external logic level. These are described in the section on triggering.

*Outputs* - Infinity provides a set of output sync signals for triggering external devices. Both pre-triggers (before the laser output pulse) and post-triggers (after the laser pulse) are available. These are described in the section on synchronization.

## **Triggering**

Firing the Infinity is a two-step process, involving two separate triggers: the first one for the lamp, the second for the Q-switch. Each trigger has two modes, internal and external, making a total of four possible triggering modes for the system. In the internal modes, Infinity's on-board electronics produce the necessary signals in accordance with the operational parameters set by the user. Internal triggering is the simplest way to run the system, since the user does not have to concern himself with the details of the triggering process.

In some applications, however, it is necessary to fire the laser on demand. Infinity offers three ways to do this:

1. Trigger the lamp with a TTL pulse and let the Infinity fire the Q-switch automatically (Lamp external, Q-switch internal).

2. Let Infinity fire the lamp automatically, but trigger the Q-switch with a TTL pulse (Lamp internal, Q-switch external).
3. Trigger both the lamp and the Q-switch with external TTL pulses (Lamp external, Q-switch external).

The choice among these three modes will depend on the application requirements. In selecting a mode, bear in mind that the time interval between the two triggers is an important parameter in system performance. When triggering the Q-switch externally, care must be taken to insure that the trigger is correctly timed with respect to the lamp pulse.

Regardless of the mode, Infinity's internal electronics have control over the *duration* of the lamp flash. The external lamp trigger simply causes the flash to begin and has no effect on its duration. The user interface allows control of the duration within certain limits, but normally it is not advisable to change it from the factory-optimized setting.

## Considerations in External Lamp Triggering

External triggering of the lamp can be used to set a custom (possibly irregular) repetition rate or to synchronize the firing of Infinity with an external device. The lamp trigger does not directly control the instant of laser emission, rather, it begins the sequence of events that ultimately results in laser output. Because the inherent jitter in these events is about 100 ns, the external lamp trigger offers relatively coarse control over laser timing.

It is not possible to exceed a repetition rate of 100 Hz in this mode. This is enforced by an on-board protection circuit that inhibits the external lamp trigger for 10 ms after each lamp flash. Any trigger pulse that arrives during this interval will be ignored. The system provides an output signal, LAMP READY, that indicates the state of this inhibit logic; this line can be monitored by external circuitry to determine when the laser is ready to fire.

Although Infinity's optical design offers uniform system performance over a wide range of repetition rates, abrupt changes in repetition rate are not recommended. Each time the repetition rate changes significantly, the temperature of critical laser components shifts to a new operating point. Until these thermal effects stabilize (in a few seconds), system performance will not be optimum; in extreme cases an excessive amount of depolarized light may result, activating the systems protection feature and shutting off the laser temporarily. Applications that encounter this problem can avoid it by firing the lamps at an approximately constant rate

(maintaining the laser in a state of thermal equilibrium), enabling and disabling the Q-switch as desired. A Q-SWITCH INHIBIT input line is supplied for this purpose.

All of Infinity's advanced control features, such as the pulse energy monitor, the depolarization monitor, and pulse energy regulation, operate independently of the lamp trigger mode. All of the TTL synchronization pulses act as described in the section on synchronization.

## **Considerations in External Q-Switch Triggering**

External triggering of the Q-switch allows precise control over the instant of laser emission: the time jitter between an external Q-switch trigger and the output laser pulse is approximately 1 ns.

In order for the system to work properly when the Q-switch is externally triggered, two timing constraints must be satisfied:

1. The lamp can be triggered either externally or internally, but it must be triggered before the Q-switch;
2. The delay between the lamp trigger and the Q-switch trigger must be close to its optimum value, which can vary slightly from one laser system to another. The wrong delay will cause low output power; jitter in the delay will cause output amplitude noise. Deviations of 1  $\mu$ s from the optimum should be considered significant.

Infinity supplies a TTL output signal that makes it simple to satisfy these timing constraints. This signal (QS READY), which is active in both internal and external lamp trigger modes, indicates that the proper interval has elapsed since the lamp trigger. The application can monitor this signal and respond by producing the Q-switch trigger, preferably within 1  $\mu$ s. Proper timing of the Q-switch is then assured without the need for external delay generators.

Within this general external triggering scheme are the two following modes:

1. External Q-switch Delayed. This mode is designed to support an application that requires overlapping the Infinity pulse with another light source, such as a sync-pumped tunable laser. Such setups typically involve arbitrary delays, both electronic and optical, that must be compensated by an adjustable delay somewhere in the system. In Infinity's External Delayed mode, the Q-switch trigger is passed through a high-resolution delay generator, programmable through the user interface in steps of approximately 0.1 ns. Furthermore,

in this mode the Q-switch trigger input is disabled until the lamps are ready (as indicated by the QS READY signal). Together these features make it quite simple to synchronize Infinity to a high repetition rate external source: trigger pulses from the external source are fed continuously to Infinity's Q-switch trigger input, and all the necessary timing and gating is handled by the on-board electronics.

2. External Q-switch Immediate. In this mode, the Q-switch fires upon receipt of the trigger. This is a high-precision, minimum-jitter trigger that allows the application to have direct control over the laser timing. This trigger is always enabled, and will cause the master laser oscillator to emit a pulse into the amplifier regardless of whether the lamps have been fired or whether the time delay between lamp and Q-switch triggers is optimal.

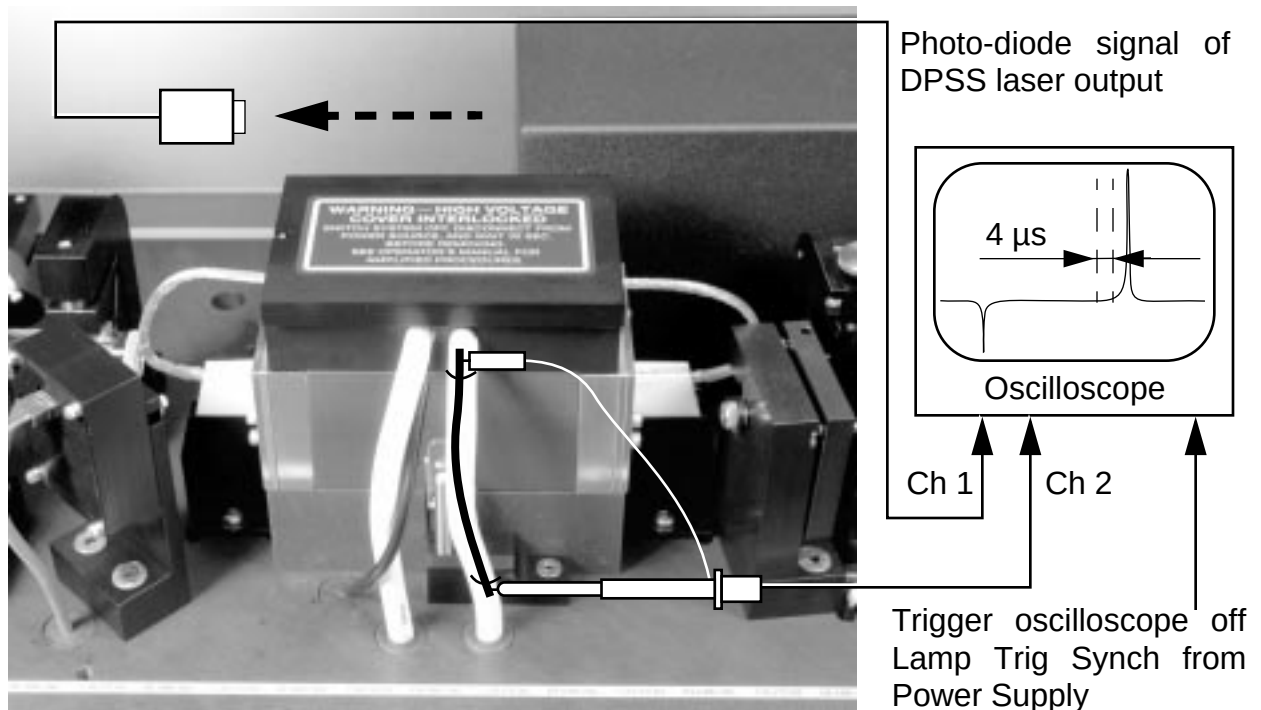
The same input BNC is used for both External Q-switch delay and External Q-switch immediate.

All of Infinity's advanced control features, including pulse energy regulation, operate independently of Q-switch triggering modes. All of the TTL Synchronization pulses act as described in the section on Synchronization, with one exception: the Variable Q-switch Delay cannot be used as a pre-trigger (i.e., cannot be programmed to produce a synch pulse that is *before* the Q-switch trigger).

## Set-Up of External Q-Switch Triggering

When the current into the flash lamp is turned off a strong electromagnetic pulse is sent through the system electronics. To avoid any degradation of system performance due to interference from this pulse it is very important that the Q-switch is timed relative the lamp current turn off with a precision greater than 0.5  $\mu$ s. In order to do this both the Q-switched laser pulse and the lamp current needs to be monitored. A convenient way of doing this is described in the following procedure.

1. Turn the key switch on the power supply to off.
2. Position a fast photodiode ( $t_{\text{rise}} < 1$  ns) in front of the DPSS laser so that it monitors the output from it. It is often easier to let the DPSS output first strike a diffuser and let the photo-diode detect scattered light. Connect the photodiode to an oscilloscope.



**Figure 3-6. Set-up to Determine Timer Settings for Lamp Pulse and Q-switch.**

3. Tape a piece of electrical wire, ~ 5 cm long, to the white flash lamp cathode lead according to Figure 4-6. Connect an oscilloscope probe to the ends of the lead as shown in Figure 4-6. Connect the probe to the oscilloscope.
4. Turn on the power supply and the computer. Run the Infinity program. Use Internal lamp and Q-switch triggering. Go to Standby mode.
5. Open the Service window. Set the lamp voltage to 360 V. Close the Shutter, click on lamp and Q-switch triggers. The lamp should now be flashing but the no laser light should exit from the system.
6. Trigger the oscilloscope of the LAMP TRIG SYNCH on the power supply.
7. Locate the signal from the probe on the oscilloscope. Use 1 M $\Omega$  termination, 50  $\mu$ s time division and 5 mV sensitivity. The signal appearing on the oscilloscope shows the time derivative of the current pulse through the flash lamp, see insert in Figure 4-6. It is created through induction in the wire taped to flash lamp lead. The inductive voltage along the wire is given by the expression:  $V = -L \frac{di}{dt}$ , where L is the inductance of the

system and  $di/dt$  the time derivative of the current through the lamp lead.

8. Locate the photo-diode signal on the oscilloscope. It should appear close to 4  $\mu$ s before the lamp turn off.
9. Note the timing relationship between the lamp turn off and the Q-switch.
10. Disable the lamp trigger in the Service window. Open the Application window.

Chose external Q-switch and if desired external lamp trigger.

Note that the duration of the lamp current pulse is always under computer control and can not be changed through the BNC connectors.

11. Connect the external trigger signals to their respective BNCs on the power supply, LAMP TRIG and QS TRIG. Note that triggering occurs on the falling edge of the applied signals.
12. Enable the lamp and Q-switch trigger signals in the Service window.
13. Position the Q-switch relative the experiment.
14. Using the signals on the oscilloscope position the lamp trigger relative the Q-switch so that the lamp turn off occurs 4  $\mu$ s after the Q-switch. Reproduce the timing sequence recorded when triggering in internal mode.
15. Go to Standby. Remove the probe and the photodiode.

## Synchronization

There are six TTL output signals from Infinity, divided into three functional groups:

*Sync signals.* Three signals are available for triggering external equipment, offering different timing and jitter characteristics. These are described in detail below.

*Ready signals.* Two signals indicate that Infinity is ready to accept external triggers, one each for the lamp and Q-switch. Named LAMP READY and QS READY, their use was described in the section on Triggering.

*Output 10 MHz clock reference.* All of Infinity's timing circuits are driven by an on-board 10 MHz clock. This clock is provided as an output allowing external counters or delay generators to Synchronize their counting rate to Infinity.

## Output Synchronizing Signals

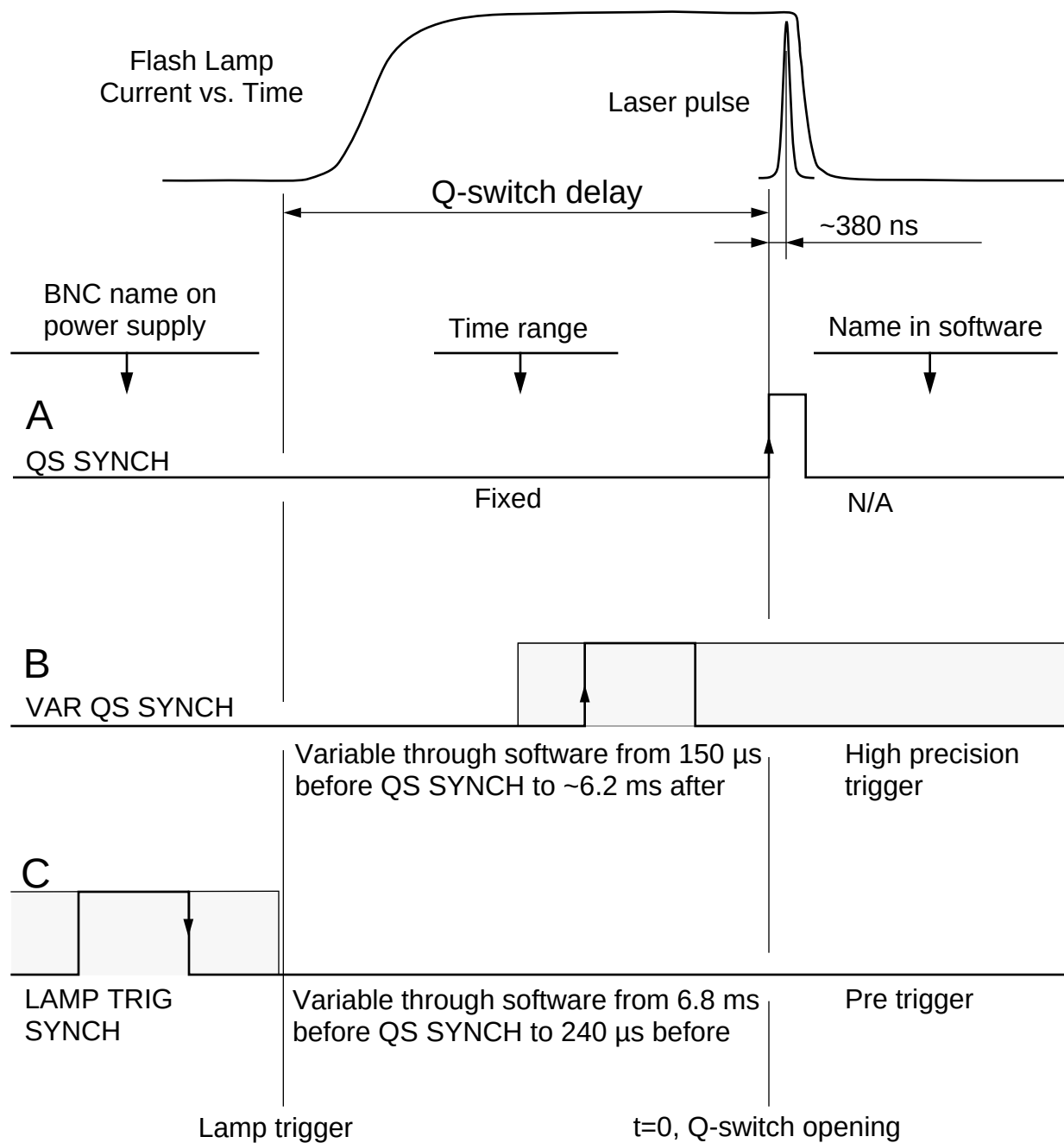
The three output triggers from Infinity serve different purposes. In their simplest terms, they provide a trigger that precedes the laser pulse by a programmable amount (LAMP TRIG SYNCH, Pre trigger in software), one that follows the laser pulse by a programmable amount (VAR QS SYNCH, High precision trigger in software), and one that is coincident with the laser pulse (QS SYNCH). The QS SYNCH pulse is a minimum jitter ( $< 0.5$  ns) trigger that permits the most precise possible synchronization of external equipment.

Note that the BNCs labeled LAMP TRIG SYNCH and VAR QS SYNCH are controlled from panels, in the Timing and triggering window, labeled Lamp pre-trigger and Q-switch sync trigger, respectively. These trigger signals and their relationship are illustrated in Figure 4-7.

The LAMP TRIG SYNCH is a pre-trigger that can be programmed to precede the Q-switch opening by between  $240\text{ }\mu\text{s}$  and  $6.77\text{ ms}$ . The resolution is one tick of the  $10\text{ MHz}$  clock ( $100\text{ ns}$ ). Equipment such as streak cameras, which typically require set-up times greater than  $100\text{ }\mu\text{s}$ , can be conveniently triggered by this signal.

The VAR QS SYNCH can be programmed as either a pre-trigger or post-trigger. As a pre-trigger, it can lead the laser pulse by approximately  $150\text{ }\mu\text{s}$ . As a post-trigger it can follow the laser pulse by up to  $6.15\text{ ms}$ . It is a high-precision trigger with a coarse and fine stage, the former having a resolution of  $100\text{ ns}$  (one tick of the  $10\text{ MHz}$  clock) and the latter  $0.5\text{ ns}$ . Each stage of the delay is independently programmable, effectively subdividing the available  $6.2\text{ ms}$  range into approximately  $10^8$  steps. This signal can be used to trigger data acquisition equipment, and is particularly useful in experiments that measure the time evolution of a sample in response to a light pulse. Such experiments can be easily automated through Infinity's programmable interfaces.

Unlike the other SYNCH signals the QS SYNCH output is not programmable, but it has the lowest timing jitter with respect to the output laser pulse. The QS SYNCH signal that appears on the power supply BNC also goes directly to the master oscillator trigger. Although laser light does not appear immediately after this trigger (there are electronic and optical delays in the laser head), the interval between the trigger and the appearance of a laser pulse is nearly constant, with a jitter of  $< 0.5\text{ ns}$ . QS SYNCH is the most precise trigger available from Infinity, and is useful for applications that demand the minimum triggering jitter.



**Figure 3-7. Relationship between Available Synchronization Signals.**



**Table 3-4. Output Signals.**

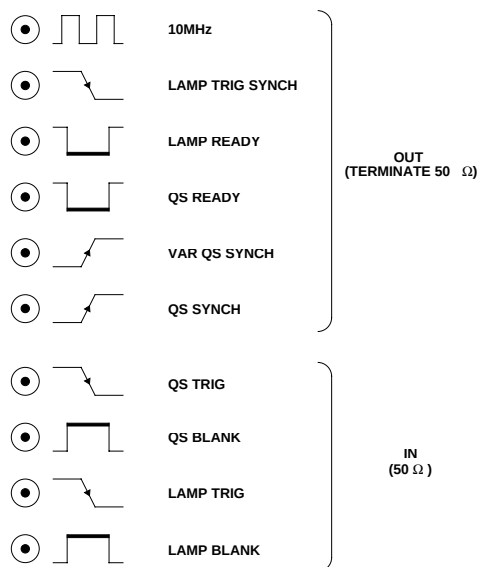
| <b>BNC</b>      | <b>FUNCTION</b>                                                                                                                                                                                                     |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10 MHz          | Master clock for Infinity timing circuits. Provided as a reference signal.                                                                                                                                          |
| LAMP TRIG SYNCH | Programmable trigger. Precedes the laser pulse by between 6.77 ms and 200 $\mu$ s. It is controlled from the lamp pre-trigger panel in the Timing and Triggering window, Figure 4-6.                                |
| LAMP READY      | Indicates by its level whether the lamp can be triggered; active only in external lamp triggering modes. Lamp triggers are disabled for 10 ms after each laser shot, enforcing a maximum repetition rate of 100 Hz. |
| QS READY        | Indicates that the Q-switch is ready to be triggered. In external Q-switch modes, the Q-switch should be fired as soon as possible after this line changes state.                                                   |
| VAR QS SYNCH    | Programmable trigger. Can precede the laser pulse by 150 $\mu$ s and follow it by 6.15 ms. It is controlled from the Q-switch sync trigger panel in the Timing and Trigger window, Figure 4-6.                      |
| QS SYNCH        | A trigger having low jitter (<0.5 ns) with respect to the laser pulse                                                                                                                                               |

**Table 3-5. Input Signals.**

| <b>BNC</b> | <b>FUNCTION</b>                               |
|------------|-----------------------------------------------|
| QS TRIG    | Fires the Q-switch on the falling edge        |
| QS BLANK   | Inhibits Q-switch firing. Active high.        |
| LAMP TRIG  | Fires the flash lamp on the falling edge      |
| LAMP BLANK | Inhibits the lamp from flashing. Active high. |

**Table 3-6. Timers and Triggering Interface.**

| PANEL                           | RANGE                                               | UNIT       | FUNCTION                                                                                                                                                                                                                 |
|---------------------------------|-----------------------------------------------------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lamp pre-trigger                | -0.2 to - 6.77                                      | ms         | Sets the time for the pre-trigger (negative time values are <i>before</i> the laser pulse)                                                                                                                               |
| High precision trigger, coarse  | -0.15 to 6.15                                       | ms         | Sets the time for the coarse stage of the programmable high-resolution trigger (negative time values are <i>before</i> the laser pulse)                                                                                  |
| High precision trigger, vernier | 0 to 255                                            | DAC counts | Sets the time for the vernier stage of the programmable high-resolution trigger. Each count represents approximately 0.1 ns; the exact calibration may vary slightly from one laser system to another.                   |
| Lamp trigger mode               | One of:<br>Internal<br>External                     |            | Selects the lamp triggering mode                                                                                                                                                                                         |
| QS Trigger mode                 | One of:<br>Internal<br>External<br>External delayed |            | Selects the Q-switch triggering mode                                                                                                                                                                                     |
| QS Trigger External delay       | 0 to 255                                            | DAC counts | Sets the delay that is applied to an input Q-switch trigger in the Q-switch External Delayed mode. Each count represents approximately 0.5 ns; the exact calibration may vary slightly from one laser system to another. |


**Figure 3-8. Layout of BNC Connectors on the Power Supply.**